Anurag Kumar Singh

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Experience

SightForge Technologies Oct 2024 – Present

Embedded Linux Engineer *Bangalore, Karnataka*

- Performed board bring-up for a custom i.MX6Q (ARM Cortex-A9) platform, including boot debugging and hardware validation.
- Developed BSPs using Yocto by creating custom meta-layers, recipes, and Linux images for embedded platforms.
- Customized Linux kernel, device tree, and U-Boot for hardware enablement, network boot, and optimized development workflows.
- Implemented hardware-accelerated media pipelines using CSI, IPU, and VPU for video capture and processing.

BoltIOT Dec 2023 – Apr 2024

Embedded Systems Intern *Remote*

- Developed multiple IoT-based projects using Arduino, ESP modules, REST APIs, and PCB prototyping.
- Integrated hardware and software components for real-time monitoring and automation applications.
- Worked on embedded programming, sensor interfacing, and communication protocols for IoT systems.

Projects

TI-DM3730-EVK-BSP | *Yocto, Linux Kernel, U-Boot, Device Tree*

- Developed a modern Yocto-based BSP for TI DM3730 (ARM Cortex-A8), replacing the legacy vendor software stack.
- Ported and customized Linux kernel and U-Boot with updated device tree support and peripheral enablement.
- Enabled NFS, NAND, and MMC boot support along with interfaces such as I2C, SPI, UART, and Ethernet.

STM32 PIO Libraries | *STM32Cube HAL, PlatformIO, Embedded C*

- Created and maintained reusable STM32 libraries for sensors and modules on PlatformIO.
- Implemented drivers for SSD1306 OLED, HC-SR04, DS1302 RTC, NEO-6M GPS, and Nokia 5110 display modules.
- Focused on modular design, portability, and easy integration across STM32-based projects.

nvim-platformio.lua | *Lua, Neovim, PlatformIO*

- Developed a Neovim plugin that provides a VS Code-like workflow for PlatformIO projects.
- Integrated build, upload, monitor, and project management features directly into Neovim.
- Maintained the project as open-source with an active user base and ongoing feature improvements.

Technical Skills

Languages: C (C89/C90), C++ (C++17), Python, Bash, Lua
Embedded Linux: Yocto Project, BSP Development, Linux Kernel, Device Tree, U-Boot, Cross Compilation
Debugging & Tools: GDB, Git, Valgrind, Perf/Ftrace (Basic), Buildroot, PlatformIO
Hardware Platforms: ARM Cortex-A (i.MX6, DM3730), ARM Cortex-M (STM32, RP2040)
Interfaces & Protocols: I2C, SPI, UART, MMC, Ethernet, CSI Camera Interface, LCD/DSS
Concepts: Embedded Systems, Linux System Programming, Boot Process, Device Drivers, Hardware Bring-up, Video Pipeline (IPU/VPU)

Education

CHANDIGARH UNIVERSITY Jul 2024 – Jun 2026

Master of Computer Applications (MCA)

Presidency College, Bangalore Oct 2021 – Jul 2024

Bachelor's Degree in Computer Applications